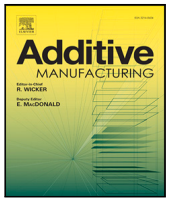
Additive Manufacturing 73 (2023) 103646



Contents lists available at ScienceDirect

Additive Manufacturing

journal homepage: www.elsevier.com/locate/addma



Research paper



Influence of layer thickness on the printability of nickel alloy 718: A systematic process optimization framework

David Shoukr [a,1], Peter Morcos [b,1], Tayler Sundermann [b], Thomas Dobrowolski [c], Chad Yates [c], Jayesh R. Jain [c], Raymundo Arróyave [a,b,d], Ibrahim Karaman [b], Alaa Elwany [a,b,*]

[a] Wm Michael Barnes'64 Department of Industrial and Systems Engineering, Texas A&M University, College Station, TX, United States of America
[b] Department of Materials Science and Engineering, Texas A&M University, College Station, TX, United States of America
[c] Baker Hughes, Houston, TX, United States of America
[d] J. Mike Walker'66 Department of Mechanical Engineering, Texas A&M University, College Station, TX, United States of America

ARTICLE INFO

Keywords:
L-PBF
Nickel alloy 718
Printability evaluation
Effect of layer thickness

ABSTRACT

Laser-powder bed fusion (L-PBF) is a metal additive manufacturing (AM) process involving layer-by-layer metal parts fabrication. It has been successfully used in the literature to manufacture nickel alloy 718 parts. However, the process parameter selection has generally been based on either intensive experimental methods or expensive computational simulations. As such, these methods test a limited region of the parameter space. Thus, hindering the evaluation of possible parameter combinations that can prove superior to those in the tested space. Moreover, most of the previous studies combine the process parameters into single metrics, e.g., linear energy density (LED) or volumetric energy density (VED). Both metrics have been shown in the literature to provide an incomplete understanding of the produced parts and inconsistent mechanical properties and microstructures.

This paper uses a systematic framework developed by the authors that covers the entire scanning speed-laser power (v-P) parameter space, estimates the maximum hatch spacing at each v-P combination, and detects the good printability region. This framework defines the good printability region as the region that is free of the main possible porosity defects, namely, lack of fusion, keyholing, and balling. Furthermore, to cover the effect of the fourth process parameter (layer thickness), the framework is applied to 60 and 90 μm layer thicknesses. This extension of the framework is performed to test the possibility of increasing the volumetric build rate (VBR) by increasing the layer thickness. Finally, the printability is validated using density and porosity measurements, while performance is evaluated using tensile tests.

Using the framework, parts with 99.99% densification were fabricated successfully for both layer thicknesses. Moreover, tensile strength values of 1059.1 and 1063.5 MPa and volumetric build rates of 7.10 and 6.92 mm^3/s were achieved for the as-printed parts for the 60 and 90 μm layer thicknesses, respectively. However, these VBR values are constrained by the conservative hatch spacing criterion used in the framework. Further optimization of the hatch spacing can provide higher VBR values for the 90 μm layer thickness.

1. Introduction

Laser powder-bed fusion (L-PBF) is one of the most common metal additive manufacturing (AM) processes, characterized by good feature resolution and surface roughness in addition to other well-established capabilities offered by AM technologies such as high levels of geometric freedom [1,2]. Although L-PBF has witnessed significant advances over the past decade, process optimization to fabricate good quality parts that meet design specifications is still a challenging task, primarily due to the large number of processing parameters that influence the outcome of the process and properties of fabricated parts [3]. Among the most commonly considered parameters that are known to have a notable influence on the bulk material performance are hatch-spacing, laser power, scanning speed, and powder-layer thickness.

Recent studies investigated the effect of different combinations of these four process parameters on the mechanical performance and microstructure of nickel alloy 718. This has been most commonly achieved through studying two aggregate metrics that combine the four parameters: linear energy density (LED) and volumetric energy

* Correspondence to: Industrial and Systems Engineering - Emerging Technologies Building, 3131 TAMU, College Station TX, 77843.
E-mail address: elwany@tamu.edu (A. Elwany).
[1] The first two authors have contributed equally to this work.

https://doi.org/10.1016/j.addma.2023.103646
Received 3 March 2023; Received in revised form 2 June 2023; Accepted 6 June 2023
Available online 12 June 2023
2214-8604/© 2023 Elsevier B.V. All rights reserved.

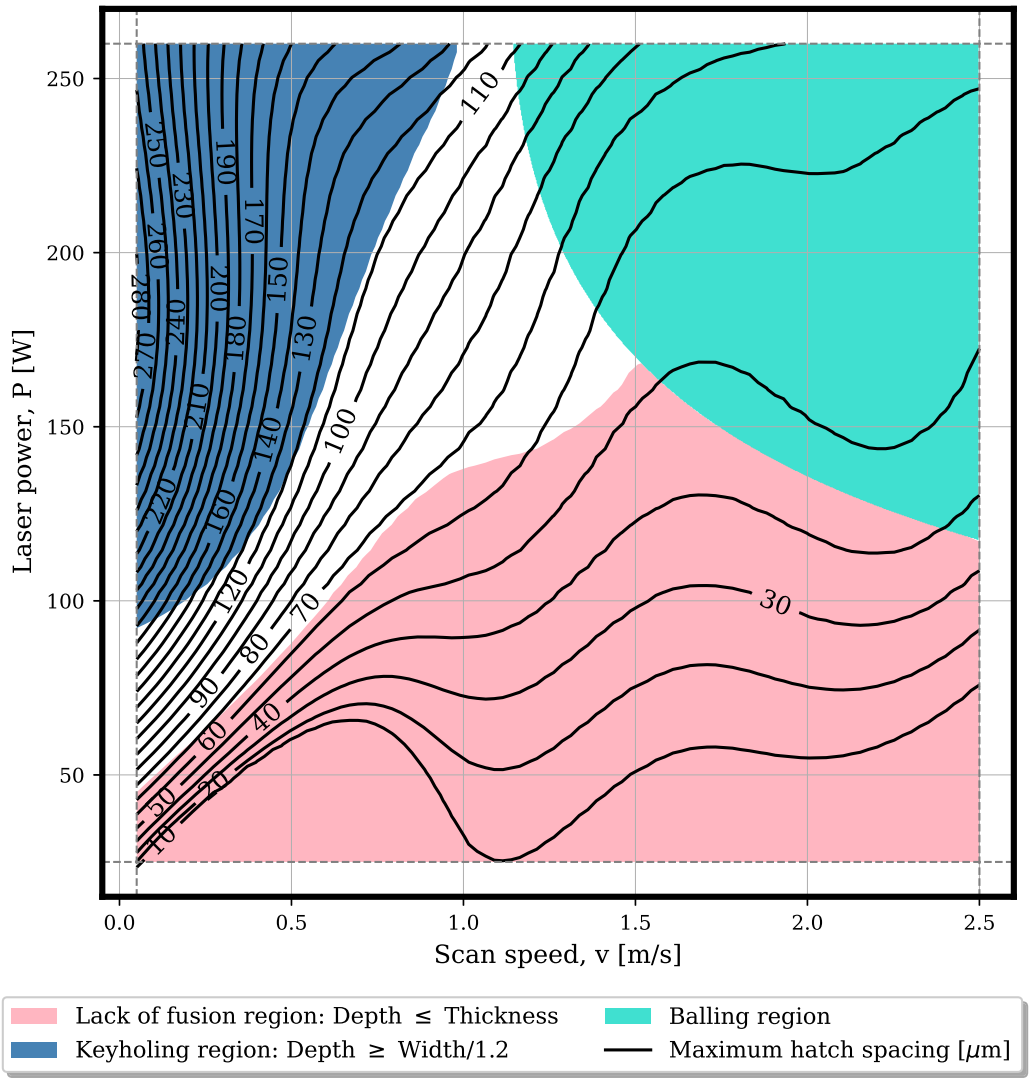


Fig. 1. A typical printability map produced using the developed framework [10].

density (VED). The LED, defined as the quotient of the laser power to the scanning speed, can be used to explain melt-pool characteristics in the L-PBF process [4,5]. Sadowski et al. [5] quantified the effects of changing LED on the consistency and uniformity of printed tracks, melt pool geometry, and defects generated during the process. They used an image processing technique on the top surfaces of single tracks to measure the apparent fill percent as a porosity metric. Other studies also established that LED considerably affects the melt-pool morphology and the microstructure of nickel alloy 718 [6], where higher values of LED result in larger melt pools [7]. The effect of LED on the microstructure was also considered in other studies. For example, Jia et al. [6] reported that microstructure goes through successive transformations: coarse columnar dendrites, then grouped dendrites, then fine dendrites, and finally cylindrical dendrites that are uniformly distributed, with increasing LED [6]. Lowering the LED also results in reduced segregation of the alloying elements, increased precipitation of the strengthening γ' and γ'' phases after aging, and lower amounts of the detrimental Laves phase [8]. LED generally provides a good explanation of melt-pool characteristics in single tracks but is limited in describing the densification of full parts since it ignores the effect of the hatch spacing and layer thickness [9].

To account for this limitation, Gu et al. [4] proposed the use of VED expressed in J/mm^3. The VED is calculated using the equation: $VED = \frac{P}{v*T*h}$, where, P is the laser power in [W], v is the scanning speed in [m/s], T is the layer thickness in [μm], and h is the hatch spacing in [mm]. Many studies used VED to take into account the effect of the four parameters on additively manufactured nickel alloy 718 [11,12]. Moussaoui et al. [13] studied the effect of VED on build properties and performance for nickel alloy 718. They fabricated cubes with 35 different combinations of laser power and scanning speed in the ranges of 150–450 W and 1.0–1.8 m/s, respectively. Only cubes with VED values in the range between 31.75 and 61.2 J/mm^3 were successfully printed. Nonetheless, VED remains to be an aggregate metric that cannot predict the effects of individual parameters on microstructure [14]. For example, Popovich et al. [15] manufactured two nickel alloy 718 parts with VED values of ≈ 59.5 J/mm^3 but with different combinations of processing parameters. Despite having the same VED, parts produced with higher laser power formed large columnar grains similar to those observed in casting processes, while parts with lower laser power showed a much finer-grained microstructure and, in turn, higher yield and ultimate tensile strengths.

To date, the majority of process parameter optimization efforts on L-PBF of nickel alloy 718 have been typically done through studying the effect of LED [5,6] and VED [4], both of which suffer some of the limitations described earlier. Moreover, all of these studies have been mainly relying on conducting large numbers of experiments that can be prohibitively costly when searching through the entire v-P design space.

Table 1
Chemical composition (in wt%) per API Standard 6ACRA [16] and observed using ICP-AES and ICP-MS techniques.

Element	Min [wt%]	Max [wt%]	Observed	Element	Min [wt%]	Max [wt%]	Observed
Ni	50.0	55.0	52.58	Si	—	0.35	0.03
Cr	17.0	21.0	19.41	P	—	0.01	<0.005
Fe	Balance		18.4	S	—	0.01	0.002
Nb	No value required		4.98	B	—	0.006	<0.005
Nb + Ta	4.87	5.2	4.98	Cu	—	0.32	0.01
Mo	2.8	3.3	3.05	Pb	—	0.001	—
Ti	0.8	1.15	0.96	Se	—	0.0005	—
Al	0.4	0.6	0.45	Bi	—	0.00005	—
C	—	0.045	0.03	Ca	—	0.003	—
Co	—	1.00	0.06	Mg	—	0.006	—
Mn	—	0.35	<0.01	W	No value required		—

The first objective of the current work is concerned with the optimization of the L-PBF printing process for nickel alloy 718 through employing a systematic framework developed earlier by the co-authors that takes into consideration the effect of individual process parameters on densification and porosity formation [10,17]. The outcome of the framework is a printability map, which identifies regions of the parameter space that correspond to porosity-inducing mechanisms (lack of fusion, keyholing, and balling). Fig. 1 shows a printability map of a martensitic steel alloy (AF9628) [10], where the contours on the map correspond to maximum hatch spacing.

The second objective is to study the feasibility of increasing productivity in the L-PBF of nickel alloy 718 through using larger layer thicknesses. In particular, different printability maps, corresponding to different layer thicknesses, are generated and then used to fabricate mechanical test samples. While the effect of layer thickness has been studied for different materials [18–26], the effect of layer thickness for nickel alloy 718 has only been studied in [19,27,28]. Sufiiarov et al. (2017) studied the use of a layer thickness of 30 μm [27] and compared it to their previous study of a 50 μm layer thickness in [29]. They found that parts produced using a higher layer thickness had on average a lower yield point while having higher elongation. Meanwhile, both layer thicknesses produced approximately the same ultimate tensile strength experimental values. Nguyen et al. [19] studied the effect of the layer thickness and found that lower layer thickness results in more dense parts and higher average yield strength (1063 MPa) and ultimate tensile strength (UTS) (745 MPa). Both studies use layer thicknesses between 20 and 50 μm with a maximum volumetric build rate (VBR) of 4.83 mm^3/s. Meanwhile, Nayak et al. [18] is the only study that investigated the effect of the different process parameters (P, v, and h) on the relative density at a high layer thickness of 100 μm but only for nickel alloy 625 [18]. Then, they compared it to data from the literature of layer thicknesses <50 μm. They found that it is possible to produce parts at higher layer thicknesses with favorable mechanical properties. Finally, Paradise et al. [28] attempted at improving the productivity by increasing the layer thickness from 30 μm to 60 μm and 80 μm and studied their effect on the surface roughness, densification, and mechanical properties. Their methodology in selecting the process parameters was solely experimental-based. Hence, they were limited by the number of experiments and could not provide an exhaustive search of the parameter space. As such, they missed feasible printing process parameter combinations that are located outside of their experimental design space. While they were able to print fully-dense parts, their maximum strengths (for as-built parts) and VBR values of ≈ 910 MPa (for **30** μm layer thickness) and 6.08 mm^3/s (for **90** μm layer thickness) are much lower than the ones achieved in the current study.

In contrast to prior studies, the proposed systematic process optimization framework ensures a thorough investigation of the process parameter space and the effect of layer thickness on productivity with reduced experimental and computational costs. The experimental methods are described in Section 2. Mechanical properties and characterization of the printed samples guided by the printability maps are reported in Section 3. Finally, the paper's conclusion and insights are drawn in Section 4.

2. Experimental and computational methods

2.1. Material

Nickel alloy 718 was developed by the International Nickel Company (InCo) in 1959 as Inconel 718. As it is a superalloy with superior high-temperature resistance and chemical properties, it is extensively used in high-temperature, load-bearing, and corrosive environments [30]. In addition, nickel alloy 718 is also used in cryogenic structures, nuclear, and oil and gas industries due to its excellent aqueous corrosion resistance and strength at ambient temperature [31].

Nickel is the main element in nickel alloy 718, which forms the austenitic face-centered cubic FCC matrix phase γ. The addition of other alloying elements provides the desired high-temperature and corrosion-resistance properties. For instance, the presence of Cr and Mo enhances the corrosion resistance of the alloy to oxidizing media and sulfur compounds, as well as pitting corrosion in many media [32]. The presence of Nb in the range of 4.5 to 6.0% at. provides age-hardenability while maintaining high-temperature stability, so the alloy can be used when creep-rupture resistance is desired [31]. Moreover, Mo, Ta, W, and Fe in the γ-matrix promote the solid-solution strengthening of the alloy, while the addition of Al, Ti, and Nb tend to form γ' and γ'' precipitates, which are responsible for the strengthening of the Ni-based superalloys.

Table 1 shows the composition of nickel alloy 718 according to API Standard 6ACRA [16] as well as the observed composition. Composition is measured using inductively-coupled plasma atomic emission spectroscopy (ICP-AES) and mass spectrometry (ICP-MS). ICP-AES is a technique that atomizes incoming specimens using super-heated plasma. It can determine concentrations of both trace and major elements. Moreover, it is able to detect most periodic table elements. Reliable results can be obtained for about 70 elements with detection limits in the order of parts per billion range [33–36].

Meanwhile, in ICP-MS, the ions from the plasma are extracted into a mass through a series of cones. Then, their mass-to-charge ratio is used to separate them. Finally, a detector receives an ion signal proportional to the concentration of the species. Compared to ICP-AES, it is much more rapid since species can be analyzed simultaneously and has a much more extensive range of elements to explore, from lithium to uranium. In addition, ICP-MS can be altered to operate slower with greater precision and can be as sensitive as parts per quadrillion [37].

The powder used in this study was gas-atomized. It had a reported particle size distribution with D_{10}, D_{50}, and D_{90} of 16.6, 33.3, and 55.7 μm, respectively. The Hall flow rate was measured according to the ASTM B23 standard [38] and found to be 16 s/50 g. Moreover, the apparent density measured according to the ASTM B212 standard [39] was 4.16 g/cm^3. SEM images of the powder showed that it is mostly spherical with some remelted and oblong particles.

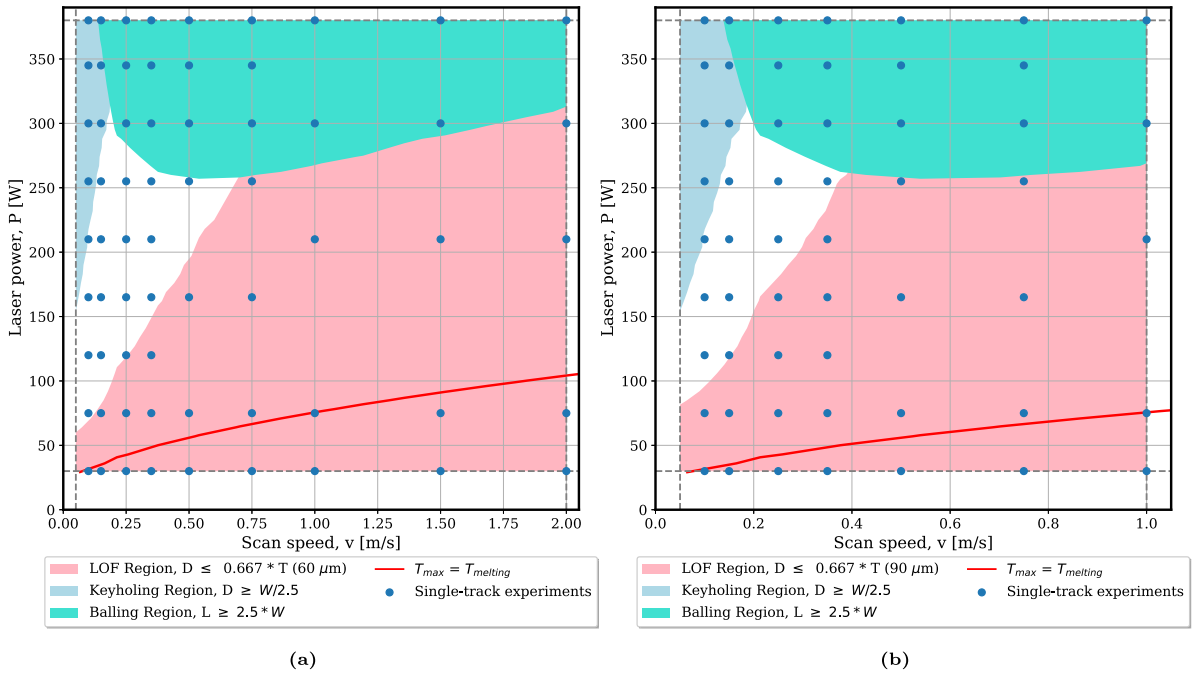


Fig. 2. Initial printability map for nickel alloy 718 for (a) 60 μm and (b) 90 μm layer thicknesses. Note: The scan speed scale for the 90 μm layer thickness is reduced to a maximum of 1 m/s for clarity.

2.2. Additive manufacturing system

The samples were manufactured on an SLM 280 L-PBF system produced by SLM® Solutions GmbH. The machine is equipped with two IPG ytterbium fiber lasers. The laser has a maximum power of 400 W, a characteristic length of emission of 1070 nm, and a maximum scanning speed of 10 m/s. The system operates under the inert atmosphere of argon gas.

2.3. Analytical model

An exhaustive analysis of the parameter space is needed to understand the printability behavior of nickel alloy 718. For this purpose, we use the Eagar-Tsai (E-T) model [40]. The E-T model generates relatively inexpensive predictions of the melt-pool dimensions. These melt-pool dimensions provide an indication of the occurrence of porosity and whether the type of porosity is lack of fusion, keyholing, or balling [41]. Compared to computationally expensive high-fidelity finite-element models, the model is relatively simple, fast, and reasonably predictive of melt-pool dimensions. Then, statistical calibration techniques are used to combine the experimental single-track results with the E-T emulator to reduce the uncertainty in the E-T model [42].

The E-T model represents the welding process as a moving normally-distributed heat source over a semi-infinite flat plate. Then, accordingly, it calculates the temperature field across that plate [40]. As such, the literature has used it to represent the melt pool in the L-PBF process [43]. The E-T model inputs are divided into two categories: heat-source parameters (namely, scan speed 'v,' laser power 'P,' and laser-beam diameter 'd'), and constant material inputs (namely, density 'ρ,' thermal conductivity 'k,' melting temperature '$T_{melting}$,' specific heat capacity 'C_p,' and absorptivity 'A').

2.4. Initial printability map

An initial printability map is generated to provide a quick understanding of the different printability regions and to help determine the location of the single tracks [17]. The initial printability map is determined by sampling the original E-T model and then determining the different regions (good, keyholing, LOF, and balling) based on the predicted melt-pool dimensions, see Fig. 2. The initial material properties, namely thermal conductivity 'k' and specific heat capacity 'C_p,' to be used in the E-T model, are taken at the melting temperature for nickel alloy 718, see Table 2. Finally, the beam diameter was taken as 100 μm as the nominal value reported by the L-PBF system vendor.

Table 2
Thermal properties of nickel alloys at melting temperature.

Thermal conductivity, k	Specific heat capacity, C_p	Absorptivity, A
30.8 W/m K [44]	775 J/kg K [44]	0.48 [45]

2.5. Single-track experiments

Single-track experiments provide an excellent representation of the expected geometry of the melt pools. They are used to revise the melt-pool geometry predictions from the E-T model and determine the melt-pool dimensional relationships for the lack of fusion, keyholing, and balling criteria.

The selection of the process parameters for the single tracks is determined such that it covers the entire possible process parameter domain. The minimum and maximum scanning speed and laser power determine the domain of the process parameters. The minimum scanning speed, v_{min}, is chosen at a very small value of 0.05 m/s to ensure a moving heat source. The maximum scanning speed, v_{max}, is determined from the literature at 2 m/s. Meanwhile, the minimum laser power, P_{min}, is selected as the smallest laser power at v_{min} that can melt the powder, with a value of 30 W. In other words, it is the intersection between the $T_{max} = T_{melting}$ (if the maximum temperature, T_{max}, is less than the melting temperature, $T_{melting}$, then no melting has occured) line and the v_{min} vertical line in Fig. 2. Finally, the maximum laser power, P_{max}, is determined as the maximum power the machine can deliver.

To ensure maximum coverage of the v-P space, uniform grid sampling is used to select the single track v and P values. The parameter space is split into three regions: a region with the highest probability, according to the initial printability map, of the existence of good prints

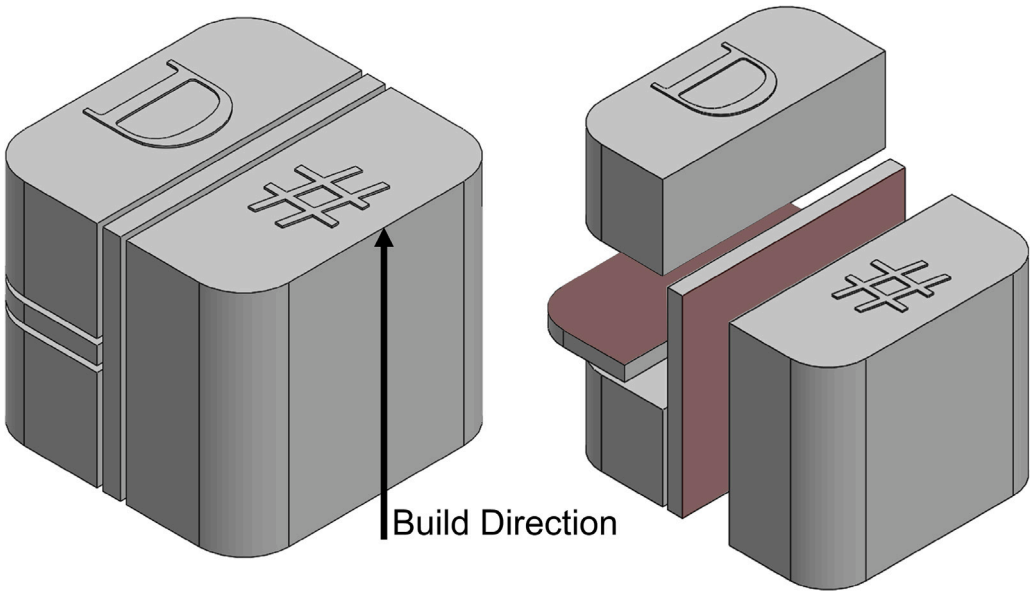


Fig. 3. The two locations where the OM images were taken with respect to the build direction are represented by the red surfaces. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Latin hypercube sample parameter bounds.

Parameter	Lower bound	Upper bound
Scanning speed, v [m/s]	0.05	2
Laser power, P [W]	30	380
Beam diameter, d [μm]	80	150
Absorptivity, A	0	1
Thermal conductivity, k [W/m K]	20	120
Specific heat capacity, C_p [J/kg K]	450	800

Table 4
Effective calibration parameter values at each layer thickness.

Calibration parameter	Layer thickness, t [μm]	
	60	90
Beam diameter, d [μm]	113.2	121.5
Thermal conductivity, k [W/m K]	32.46	52.38
Absorptivity, A	0.73	0.48
Specific heat capacity, C_p [J/kg K]	537.5	472.2

in which 36 equally spaced tracks were selected, a second region with a lower probability with 14 tracks, and a third region with mostly LOF and balling regions with 15 tracks, see Fig. 2. The single tracks are also spread such that the extremities of the process space are included. This is important for the calibration step described in Section 2.6 to avoid extrapolation when fitting the Gaussian process regression model.

After performing the single tracks for the 60 μm layer thickness, it was decided that the domain for the 90 μm layer thickness be reduced to a v_{max} of 1 m/s. The three grid distributions were then split into 36, 14, and 5 points grids for a total of 55 single tracks.

For each layer thickness, 10 mm long single tracks were printed 1 mm apart from each other on a previously printed platform covered by a single layer of powder. The single tracks were printed in vectors perpendicular to the vectors of the previously printed platform for better measurability of the melt-pool dimensions. The platform vectors were printed uni-directional against the gas flow. Three cross-sections were cut at quarter, half, and three-quarters of each track's length using electric discharge machining (wire EDM). Afterward, these cross-sections were polished down to 0.25 μm, and Kalling's reagent No. 2 (specifically, 5 g $CuCl_2$, 100 mL ethanol, and 100 mL HCl) was applied to etch the surface of the single-track cross-sections for three rounds of five second etch time, then flushing with water. Each track's width and depth were measured thrice at each cross-section, then the averages were used as the experimental results.

2.6. Model calibration

The model is calibrated according to the Kennedy and O'Hagan two-stage framework [46]. The first of the calibration framework stages is to build an inexpensive Gaussian process surrogate of the E-T model. A sample of 500 points was queried from the original E-T model to develop this surrogate using the Latin hypercube (LH) sampling method. The chosen sample parameters are divided into three categories: (1) control, (2) calibration, and (3) fixed parameters. Control and calibration parameters are varied according to Table 3. For details of the selection of these parameter bounds, please refer to [10,17]. Meanwhile, fixed parameters, i.e., melting temperature '$T_{melting}$' and density 'ρ,' were kept constant at 1,609 K, and 8,193 kg/m^3, respectively.

In the second stage, the single tracks' melt-pool width and depth measurements are used to calibrate the surrogate model. As part of the calibration process, hyperparameters of the surrogate model and a discrepancy function are optimized to estimate calibration parameters. The mean absolute prediction error (MAPE) is reported as the performance metric of the calibrated model. The MAPE for width is 6.71%, and for depth is 13.31% for the 60 μm layer thickness and 5.85% and 19.38% for the 90 μm layer thickness. The estimated melt-pool width and depth values are used to build the LOF and keyholing boundaries in the printability map as well as calculate the maximum hatch spacing contours. Table 4 shows the calibration parameter values for the 60 and 90 μm layer thicknesses based on the performed calibration. It is worth mentioning that these values represent the effective parameter values for each quantity and not the exact parameter estimates. This is because of the inclusion of the discrepancy function in the calibration model [42].

2.7. Printability maps

A fine grid from the calibrated E-T model is sampled to construct the printability map. The grid consists of 100*100 = 10,000 points equally spaced in the v and P directions. For each point, the melt-pool depth and width values are predicted. Based on these values, the different criteria for the LOF and keyholing are plotted. The boundary of the

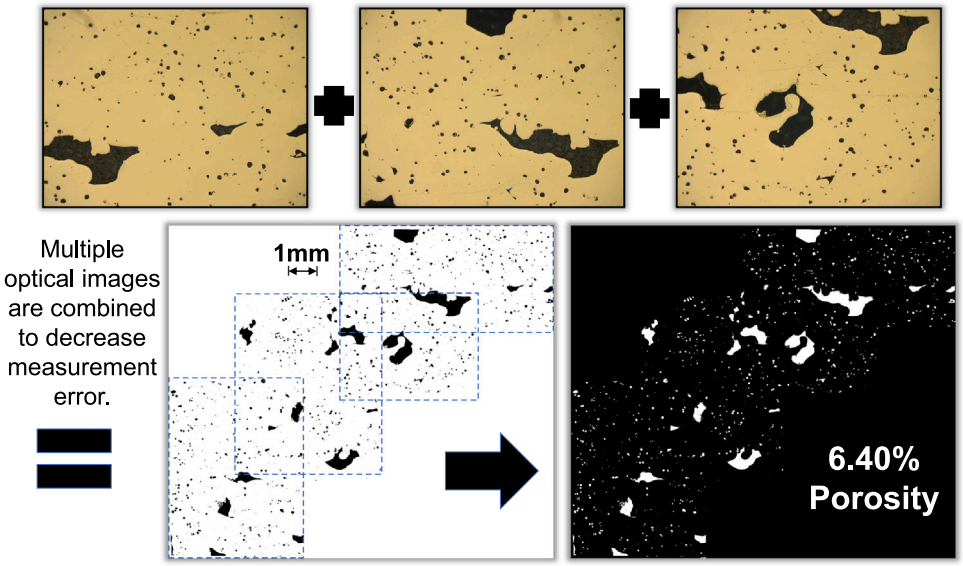


Fig. 4. Example of combining optical microscopy images at the same location.

LOF region is determined by the points in the grid that have a depth, $D \leq 0.667$*the layer thickness, T. Meanwhile, the keyholing region is determined by the melt-pool depth-to-width ratio. The exact criterion value for keyholing is chosen to fit best the single-track experiments classified as keyholing. The balling criterion depends on the melt-pool width-to-length ratio. As the melt-pool length cannot be typically measured (because of the overlapping melt pools), the effective values of the calibration parameters, obtained in Section 2.6, are used to obtain a coarser grid from the original E-T model to predict the melt-pool length values. Then the criterion value for balling is determined based on the single-track balling points.

2.8. Maximum hatch spacing criterion

The maximum hatch spacing, h_{max}, represents a conservative upper bound on the hatch spacing, h, for each v-P combination that will result in adequate fusion between successive layers, and successive tracks within the same layer. This value is calculated using a geometric criterion developed and described in [10]. This geometric criterion is based on two assumptions: (a) the melt pool has a parabolic shape both above and below the substrate's surface, and (b) the melt-pool height is equal to the layer thickness. According to this geometric criterion, the overlapping between consecutive hatch lines (the combination of overlapping height, OH, and depth, OD) is at least equal to the layer thickness. This ensures full fusion within the same layer. Moreover, assuming we are printing layer i and the subsequent layer is i+1, the overlapping depth for the subsequent layer, OD_{i+1} is equal to that of the current layer OD_i, the resultant between-layer overlapping (OD_{i+1}+ OH_i) is also at least equal to the layer thickness. Thus, ensuring full inter-layer fusion and eliminating any lack-of-fusion porosity that might occur due to an excessive hatch spacing value. The resulting maximum hatch spacing becomes

$$h_{max} = W\sqrt{1 - \frac{T}{(T+D)}}$$

where, T is the layer thickness, and, W and D are the predicted melt-pool width and depth corresponding to each v-P combination.

2.9. Experimental methods

Two types of testing are performed to validate each printability map's printable region: densification/porosity and tension tests. Density samples are first selected from the printability map to cover the entire printability region. These 10*10*10 mm^3 density coupons are then tested for density and porosity as described in Section 2.9.1. Then the process parameter combinations that produce the highest density levels are selected to be tested in tension [17], as described in Section 2.9.2.

2.9.1. Density and porosity measurement methods

Two techniques were used to test the relative density of the printed coupons: the Archimedes method performed according to the ASTM B962-15 standard [47] and optical microscopy (OM) analysis. The latter provides further insights into the pores' shape, size, and distribution. The Archimedes density is measured in ethanol to obtain submerged weight and then compared to a reference cube density made from the wrought nickel alloy 718 at 8,267 kg/m^3. For the OM analysis, images were taken in vertical and horizontal cross-sections; see the red surfaces in Fig. 3. Multiple optical images were taken at each location and then combined to decrease the measurement error, as shown in Fig. 4. The images are then processed using the ImageJ® software [48] to calculate the percent porosity at each location. The OM relative density is (100-porosity)%.

2.9.2. Mechanical testing

The tension test was used to determine the yield strength, ultimate tensile strength, elongation at fracture, and toughness for select samples with the highest densification. The tensile samples used in the experimentation are sub-size samples. The samples are of the pin-loaded tension test specimen described in Fig. 5a of ASTM standard E8/E8M [49]. Its dimensions are shown in the figure, with a thickness of 0.75 mm so as not to overload the hardened steel pins used to hold the specimen in place during testing. Samples were cut using wire EDM from the larger sample into multiple slices, shown in Fig. 5b, and holes were inserted using plunge EDM. Specimen 1 starts from the bottom of the sample and proceeds up in ascending order. The specimens were tested at a constant cross-head speed of 0.015 mm/min using an MTS 8 mm Epsilon hydraulic extensometer with a 250 kN load cell. The tests utilized a ceramic extensometer (MTS model 632.53E−14) for strain measurement.

3. Results and discussion

3.1. Printability maps

The regions of the printability map (good, balling, keyholing, and LOF) are determined with the help of the single-track classifications,

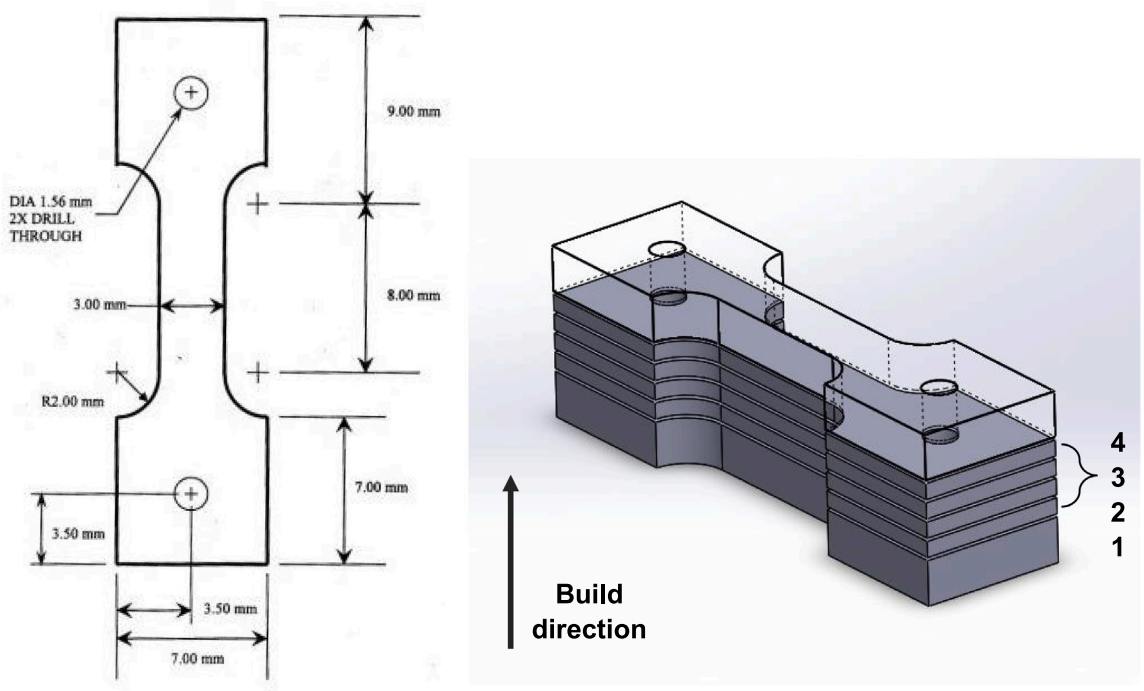


Fig. 5. Description of the used sub-sized tension samples showing (a) the dimensions of the sample, and (b) sample numbering with respect to building direction.

see Fig. 6. For both layer thicknesses, the LOF region is where the points with melt-pool depths less than or equal to two-thirds the layer thickness [17]. For the 60 μm layer thickness, the keyholing region was chosen such that all the points within satisfy the criterion depth (D) ≥ width (W)/1.7. While, for the 90 μm layer thickness, it was D ≥ W/1.5. These criteria values are chosen such that all the keyholing single tracks are included in the region. Meanwhile, the balling regions were determined such that they satisfy the criterion length (L) ≥ 4*W and L ≥ 1.95*W for the 60 and 90 μm layer thicknesses, respectively. The selection of these criteria values is such that the single-track balling points are included. While the current criteria are satisfactory for the current study purposes, further research is needed to provide a more consistent process for establishing more general criteria. It is worth mentioning here that there is a great overlap between the LOF and the balling regions for both layer thicknesses.

After the defect regions are determined, the printability region is basically the defect-free region. Using the maximum hatch spacing criterion described in Section 2.8, the maximum hatch spacing contours are superimposed on the printability map regions; see Fig. 7. Hence, the printability map, corresponding to each layer thickness, has the four process parameters (scanning speed, laser power, hatch spacing, and layer thickness). For each v-P combination in the good printability region, one can select a suitable hatch spacing value based on the hatch-spacing contours.

3.2. Density and volumetric build rate results

To validate the printability maps, density cubes are selected, covering the v-P space of the good printability region. As the probability of LOF occurrence in the 90 μm layer thickness is much greater, the good printability region is much smaller. As a result, only nine density cubes are selected for the 90 μm layer thickness, see Figs. 8(b) & 9(b), while 15 cubes were chosen for the 60 μm layer thickness, see Figs. 8(a) & 9(a). Both the Archimedes relative density and the OM porosity tests are applied to density cubes. The OM and Archimedes density values are tabulated in Tables 5 & 6 for the 60 and 90 μm layer thicknesses, respectively. It can be seen that the low speed-low power region is providing much higher porosity, especially with the 60 μm layer thickness case. This is mainly due to the high spatter that occurs while printing with such low-speed [50]. For both layer thicknesses, OM relative density values of up to 99.99% are achieved. That means that a porosity of ≤ 0.01% is detected in the density cube. Meanwhile, the Archimedes density values are relatively smaller than the OM densities. This can be attributed to the high reference density value being used of 8,267 kg/m^3. Nevertheless, there is a consistency between the two measurements, as the Archimedes density is showing values of 99+% for the corresponding cubes with 99+% OM density.

Table 5
Density and volumetric build rate results for the selected cubes for the 60 μm layer thickness.

Sample	v [m/s]	P [W]	h_{max} [μm]	VBR [mm^3/s]	OM density [%]	Archimedes density [%]
1	0.15	125	176	1.585	93.93	93.95
2	0.15	175	243	2.186	96.60	94.23
3	0.38	150	111	2.532	98.09	98.22
4	0.38	200	151	3.433	98.98	98.26
5	0.55	225	127	4.177	99.94	99.37
6	0.55	300	153	5.054	99.83	99.34
7	0.75	250	104	4.683	99.98	99.45
8	0.75	300	122	5.512	99.99	99.48
9	0.75	370	146	6.570	99.96	99.45
10	0.75	380	158	7.102	99.96	99.53
11	0.80	350	131	6.308	99.99	99.47
12	1.00	250	76	4.585	99.93	99.44
13	1.00	315	96	5.736	99.97	99.49
14	1.00	370	109	6.536	99.96	99.45
15	1.20	370	87	6.240	99.97	99.48

The main reason for testing the printability of nickel alloy 718 at different layer thicknesses is to increase the production rate. Hence, the printability maps with the VBR contours superimposed to it are plotted in Fig. 10. The VBR for the density cubes is also found in Tables 5 and 6. It can be seen that VBRs up to 7.102 and 6.922 mm^3/s can be achieved with OM density of 99.9+% for both the 60 and 90 μm layer thicknesses. Even though the layer thickness has been increased,

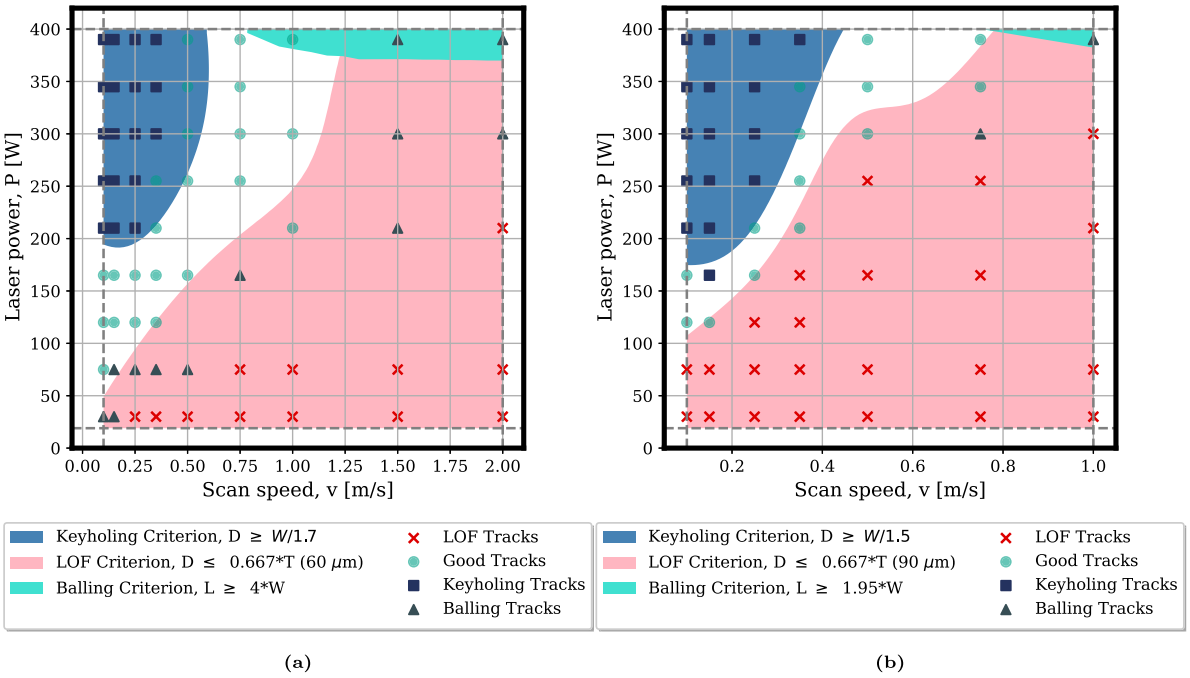


Fig. 6. Printability map of nickel alloy 718 using (a) 60 μm and (b) 90 μm layer thicknesses with single track classification superimposed.

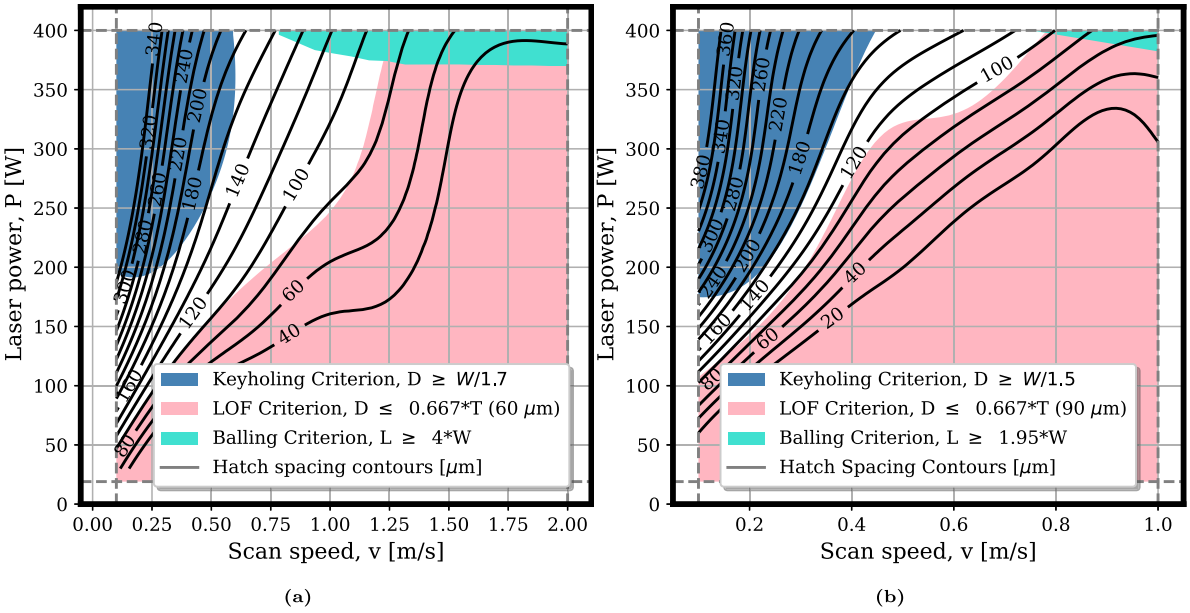


Fig. 7. Printability map of nickel alloy 718 using (a) 60 μm and (b) 90 μm layer thicknesses.

the necessary combination of scanning speed, laser power, and hatch spacing needed to achieve complete and homogeneous melting of nickel alloy 718 results in a maximum VBR equal to the maximum VBR of the 60 μm layer thickness. Nevertheless, both VBR values represent an improvement of more than 40% higher than the reported maximum VBR in the literature of 4.83 mm^3/s for 50 μm layer thickness [19], 2.16 mm^3/s for 60 μm layer thickness and 15% more than the 6.08 mm^3/s for 80 μm layer thickness [28].

3.3. Tensile testing

The six processing parameters with the highest OM density values for each layer thickness were then tested under tension. For each parameter combination, three sub-sized tension specimens were machined using wire-EDM with the wire cut parallel to the build direction and the gauge length at the center of the fabricated part; see Fig. 5. The resulting stress–strain diagrams are shown in Fig. 11. Moreover, the complete results for the tension samples are shown in Tables 7 & 8.

On average, the UTS and yield strength of the 60 μm layer thickness seems to be higher than the 90 μm one, with averages of UTS = 1047.3 MPa and σ_y = 754.1 MPa, and UTS = 1020.1 MPa and σ_y = 647.1 MPa, for the 60 and 90 μm layer thicknesses, respectively. The observed UTS for both layer thicknesses are consistent with the reported in [27] of 1051 MPa and in [19] of 1050 MPa for 50 μm layer thickness. Moreover, these UTS results are much higher than the UTS values of ≈ 850, and 900 MPa reported in [28] for the as-built specimens for the 60 and 80 μm layer thicknesses, respectively. Meanwhile, the yield strength σ_y

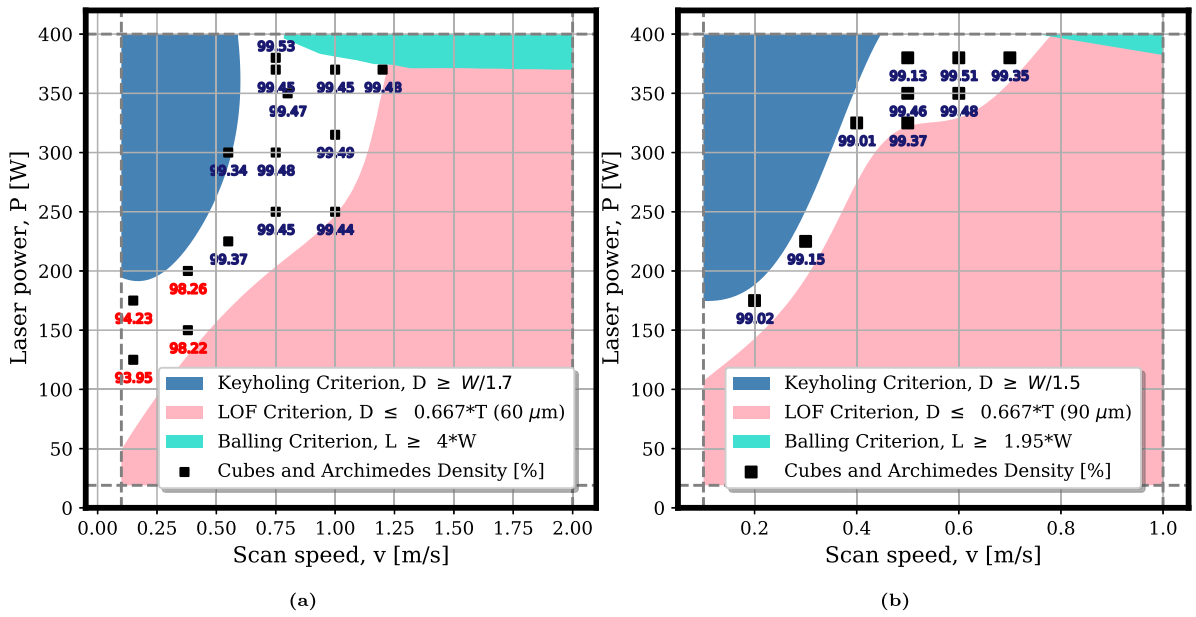


Fig. 8. Relative Archimedes density of the selected cubes for (a) 60 μm and (b) 90 μm layer thicknesses.

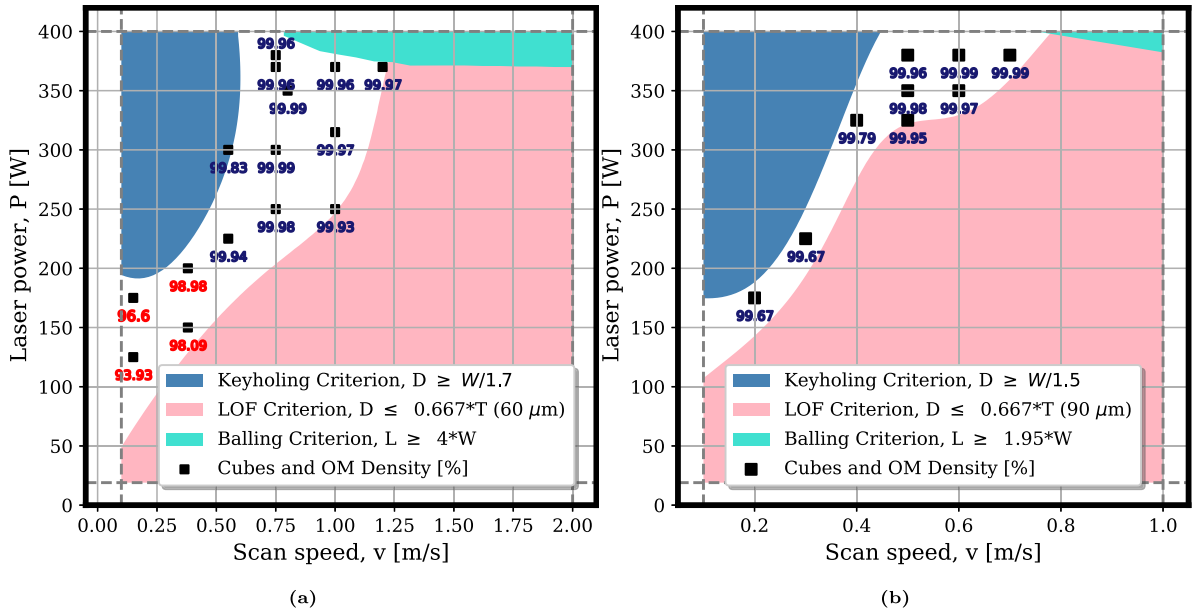


Fig. 9. Relative density using optical microscopy imaging of the selected cubes for (a) 60 μm and 90 μm layer thicknesses.

Table 6
Density and volumetric build rate results for the selected cubes for the 90 μm layer thickness.

Sample	v [m/s]	P [W]	h_{max} [μm]	VBR [mm^3/s]	OM density [%]	Archimedes density [%]
1	0.2	175	149	2.676	99.67	99.02
2	0.3	225	135	3.651	99.67	99.15
3	0.4	325	144	5.198	99.79	99.01
4	0.5	325	109	4.923	99.95	99.37
5	0.5	350	127	5.693	99.98	99.46
6	0.5	380	146	6.577	99.96	99.13
7	0.6	350	105	5.675	99.97	99.48
8	0.6	380	128	6.922	99.99	99.51
9	0.7	380	108	6.810	99.99	99.35

for both layer thicknesses was consistent with the reported 646 MPa in [27] and 732 MPa in [19] for the 50 μm layer thickness.

The average toughness and elongation at fracture for the 60 μm are slightly higher than that of the 90 μm ones. The average toughness values are 173.9 and 161.4 kJ/mm^3, for the 60 and 90 μm layer thicknesses, respectively. While the average elongation at fracture values are 18.1% and 17.7%, respectively. It is important to note that elongation at fracture values reported in the literature are all for standardized tensile test samples. Moreover, the elongation at fracture values highly depend on the test sample geometry [51]. Hence, their comparison with the observed sub-sized sample results is not feasible.

Considering the variability in the quantity of interest (QoI) values, an independent sample t-test is implemented to determine the effect of the layer thickness on the tensile properties. For all QoI's, the null and alternative hypotheses are

$$H_0 : QoI_{60} = QoI_{90}$$

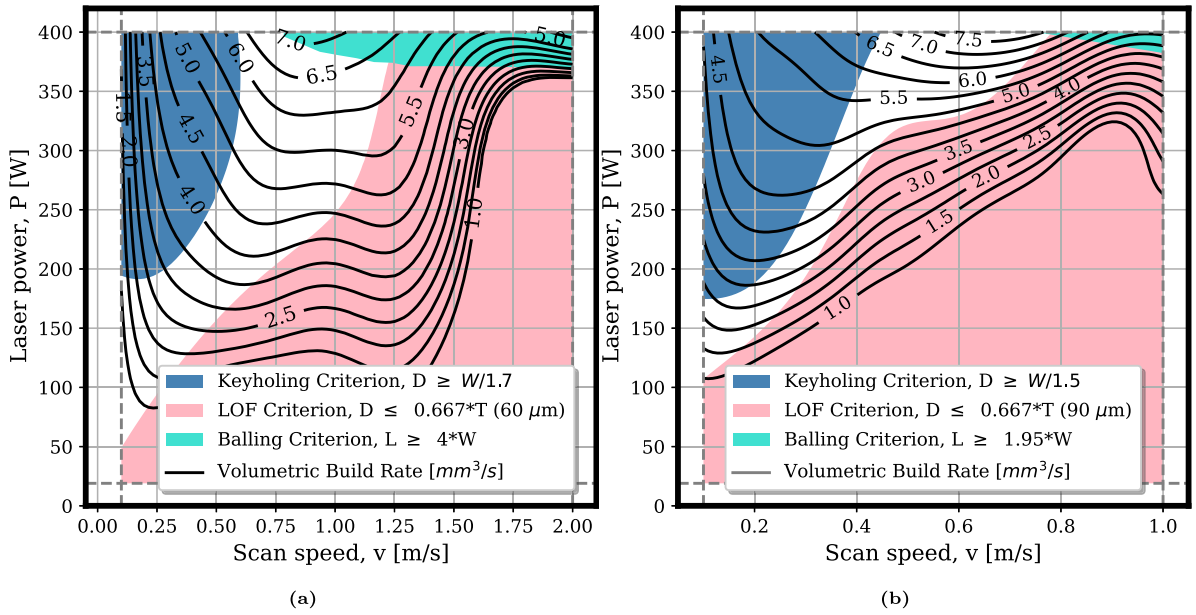


Fig. 10. Printability maps for (a) 60 μm and (b) 90 μm layer thicknesses with the volumetric build rate contours superimposed.

Table 7
Tension test results for the 60 μm layer thickness samples.

Sample	v [m/s]	P [W]	h_{max} [μm]	VBR [mm³/s]	Density OM [%]	σ_y [MPa]	UTS [MPa]	Toughness [kJ/mm³]	Elong. at **fract. [%]**
1	0.55	225	115	3.795	99.94	749.7 ± 36.2	1059.1 ± 15.8	184.8 ± 5.0	19.2 ± 0.5
2	0.75	250	105	4.683	99.98	779.7 ± 33.2	1059.0 ± 40.3	173.7 ± 12.0	17.8 ± 0.7
3	0.75	300	122	5.512	99.99	762.7 ± 9.2	1036.6 ± 11.0	168.9 ± 1.6	17.7 ± 0.2
4	0.80	350	131	6.308	99.99	745.7 ± 36.3	1032.3 ± 36.4	168.3 ± 4.5	17.8 ± 0.6
5	1.20	370	87	6.240	99.99	746.0 ± 2.6	1059.3 ± 6.6	173.0 ± 7.0	17.8 ± 0.6
6	1.00	370	109	6.536	99.96	741.0 ± 14.5	1037.5 ± 14.5	174.5 ± 0.8	18.3 ± 0.2

Table 8
Tension test results for the 90 μm layer thickness samples.

Sample	v [m/s]	P [W]	h_{max} [μm]	VBR [mm³/s]	Density OM [%]	σ_y [MPa]	UTS [MPa]	Toughness [kJ/mm³]	Elong. at fract. [%]
1	0.20	175	110	1.980	99.67	657.7 ± 50.5	1046.6 ± 24.2	179.3 ± 8.4	19.2 ± 0.9
2	0.30	225	110	2.970	99.67	716.7 ± 17.2	1025.7 ± 39.5	105.5 ± 35.7	11.5 ± 3.4
3	0.40	325	110	3.960	99.79	680.3 ± 60.5	1063.5 ± 12.6	160.9 ± 4.0	17.1 ± 0.5
4	0.50	350	127	5.693	99.98	596.3 ± 83.7	1012.7 ± 49.2	184.5 ± 16.0	20.4 ± 1.2
5	0.60	380	128	6.922	99.99	634.3 ± 82.6	1000.9 ± 11.6	166.5 ± 18.0	18.6 ± 1.5
6	0.70	380	108	6.810	99.99	597.3 ± 100.3	977.0 ± 37.6	171.5 ± 15.1	19.6 ± 1.2

Table 9
T-test results for the tensile samples.

QoI	Mean ± std		T-value	P-value
	60 μm	90 μm		
UTS [MPa]	1047.3 ± 27.2	1021.1 ± 43.1	2.12	0.021
Yield strength [MPa]	754.1 ± 25.0	647.1 ± 72.5	5.75	<0.001
Elongation at fracture [%]	18.1 ± 0.7	17.7 ± 3.4	0.45	0.323
Toughness [kJ/mm³]	173.9 ± 8.4	161.4 ± 32.34	1.54	0.066

$$H_1 : QoI_{60} > QoI_{90}$$

where, QoI_{60} and QoI_{90} are the mean of the QoI for the 60 and 90 μm layer thicknesses, respectively. The test results indeed show that, at $\alpha = 0.05$ level of significance, the UTS and σ_y for the 60 μm layer thickness are on average greater than these for the 90 μm ones. Meanwhile, there was no sufficient statistical evidence to reject the null hypothesis for the elongation at fracture and toughness. Thus, they are consistent for both layer thicknesses. The detailed t-test results are shown in Table 9.

Studying the tensile results, non-uniformities were observed in the performance of a number of samples. Most notable is the apparent premature failure in samples D2-2 and D2-3 in the 90 μm regime; premature failure here refers to fracture prior to observable necking and significantly reduced elongation as compared to other samples from the same specimen and 90 μm samples as a whole. Part porosity, microstructure, and fracture surfaces were all examined in an attempt to understand the mechanics of the event. Investigated sample locations in the printability maps are included in Fig. 12. It should be noted that all of these parts are in the "good" printability region as shown in the printability maps, Fig. 9. Fracture surfaces examined under back-scattered electrons (BSE) using a bench-top Phenom scanning-electron microscopy (SEM) are detailed in Fig. 13, in the interest of space and image detail; the best-captured surface is used in the figure.

The fracture surfaces of materials with high elongation show indications of ductile fracture, and high magnification images (not included for brevity), show dimpling consistent with the literature for as-built

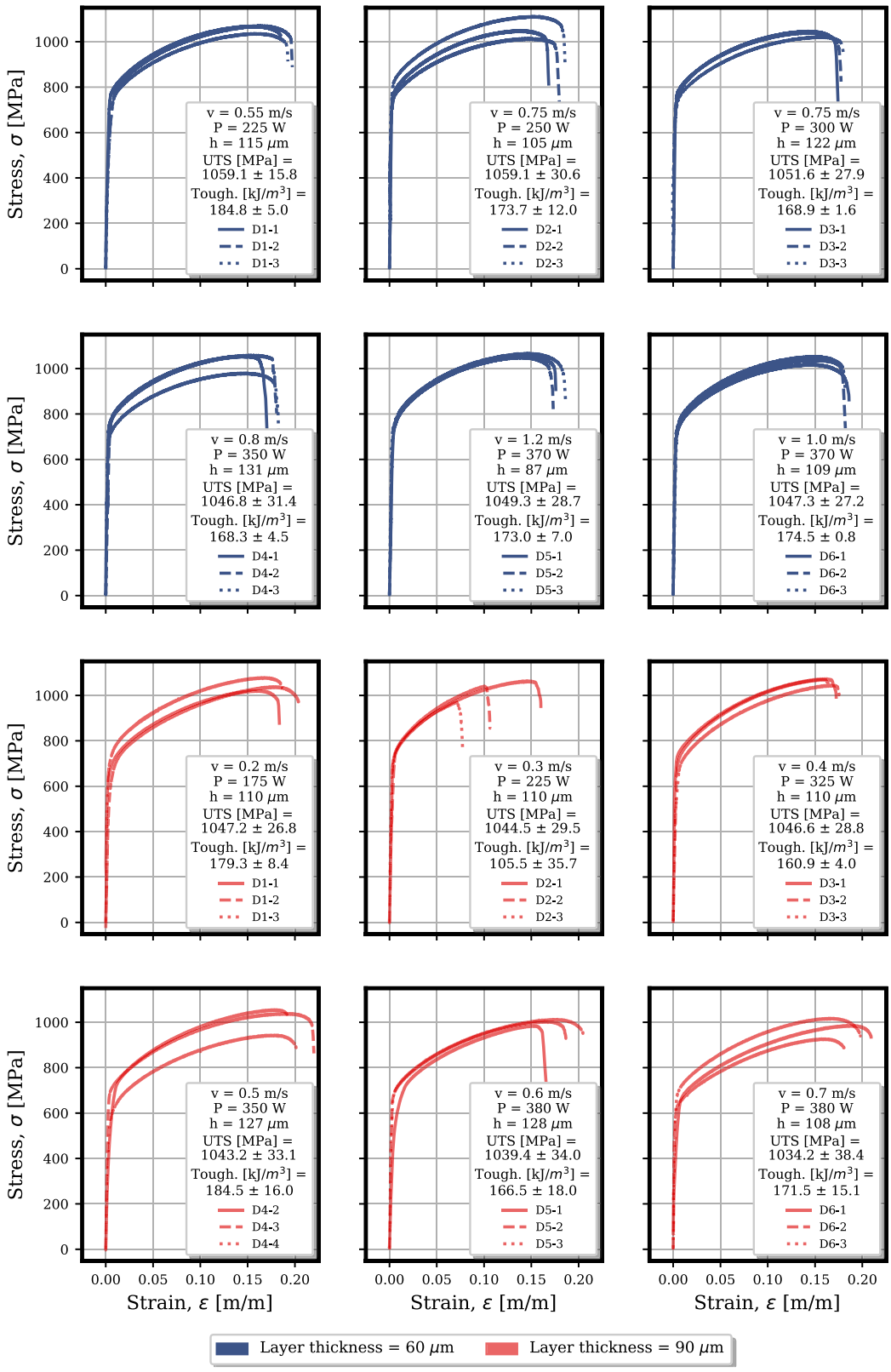


Fig. 11. Tensile test results for the selected process parameters for layer thickness = 60 μm (blue) and 90 μm (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

nickel alloy 718 L-PBF specimens [52]. The premature fracturing samples D2-2 and D2-3 both revealed large cavernous pores inconsistent with keyhole porosity; the partial fusing of powder viewable in the SEM images indicates instead a lack of fusion, see Figs. 13(c) and 13(d). However, as noted in the printability map, sample D2 is supposed to be in the good printing region, additionally, the mechanism for the lack of layer cohesion was not obvious in our examination. It is fairly obvious though that the lack of fusion increased with an increasing distance

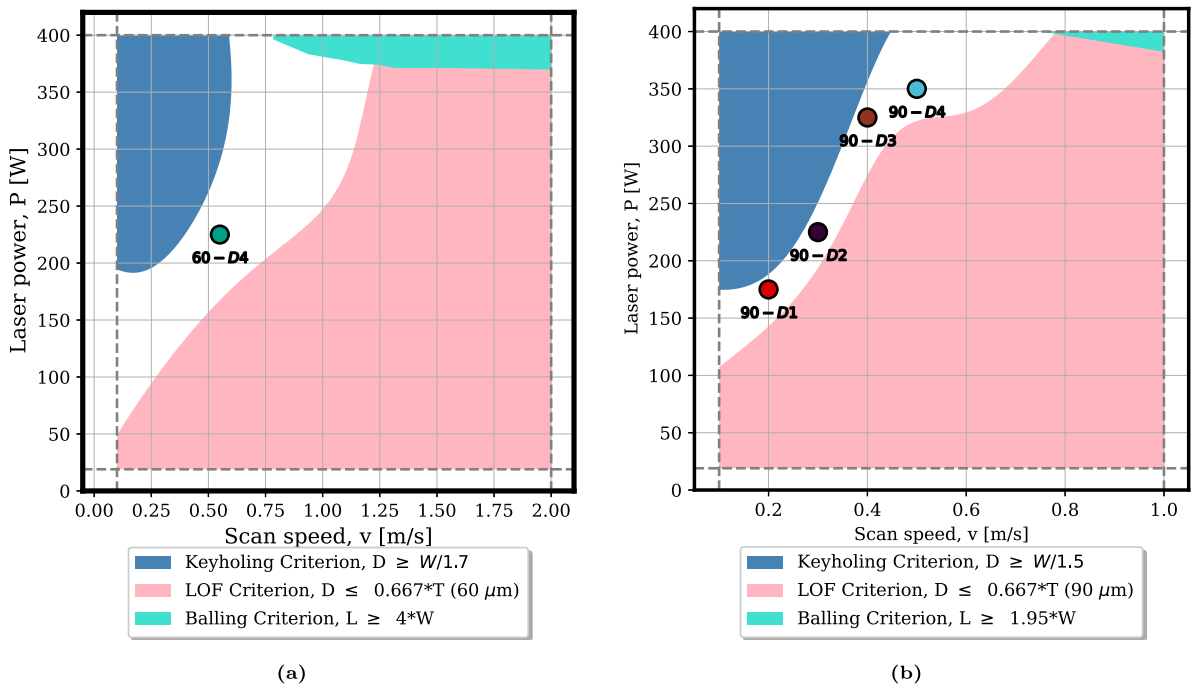


Fig. 12. Printability maps for (a) 60 μm and (b) 90 μm layer thicknesses showing the points selected to explain the non-conforming samples.

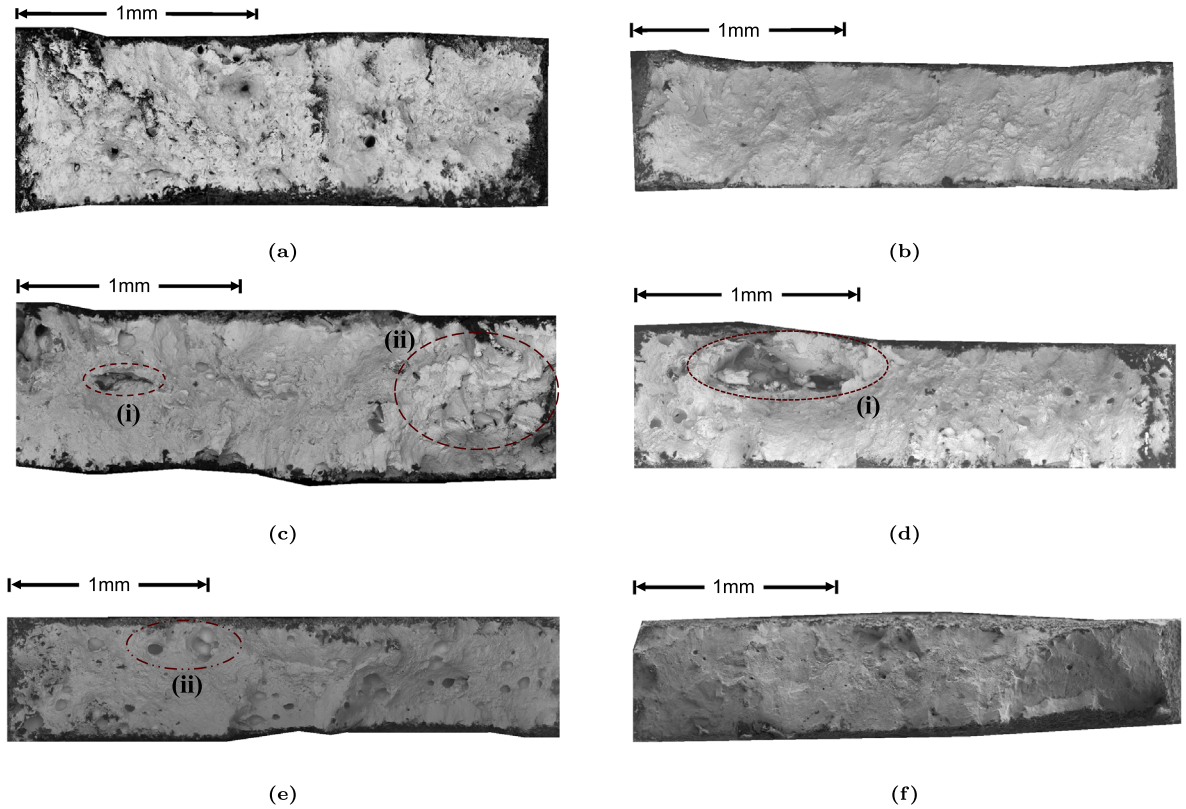


Fig. 13. Fracture surfaces of selected tensile samples for both 60 μm and 90 μm layer thickness; (a), (b), (e), and (f) are representative of the average fracture surface for given build parameters whereas (c) and (d) show the reasons for premature fracture as lack of fusion (i) as compared to keyholing porosity (ii) which seems to be non-contributory to fracture initiation.
Note: each image has an individual scale bar.

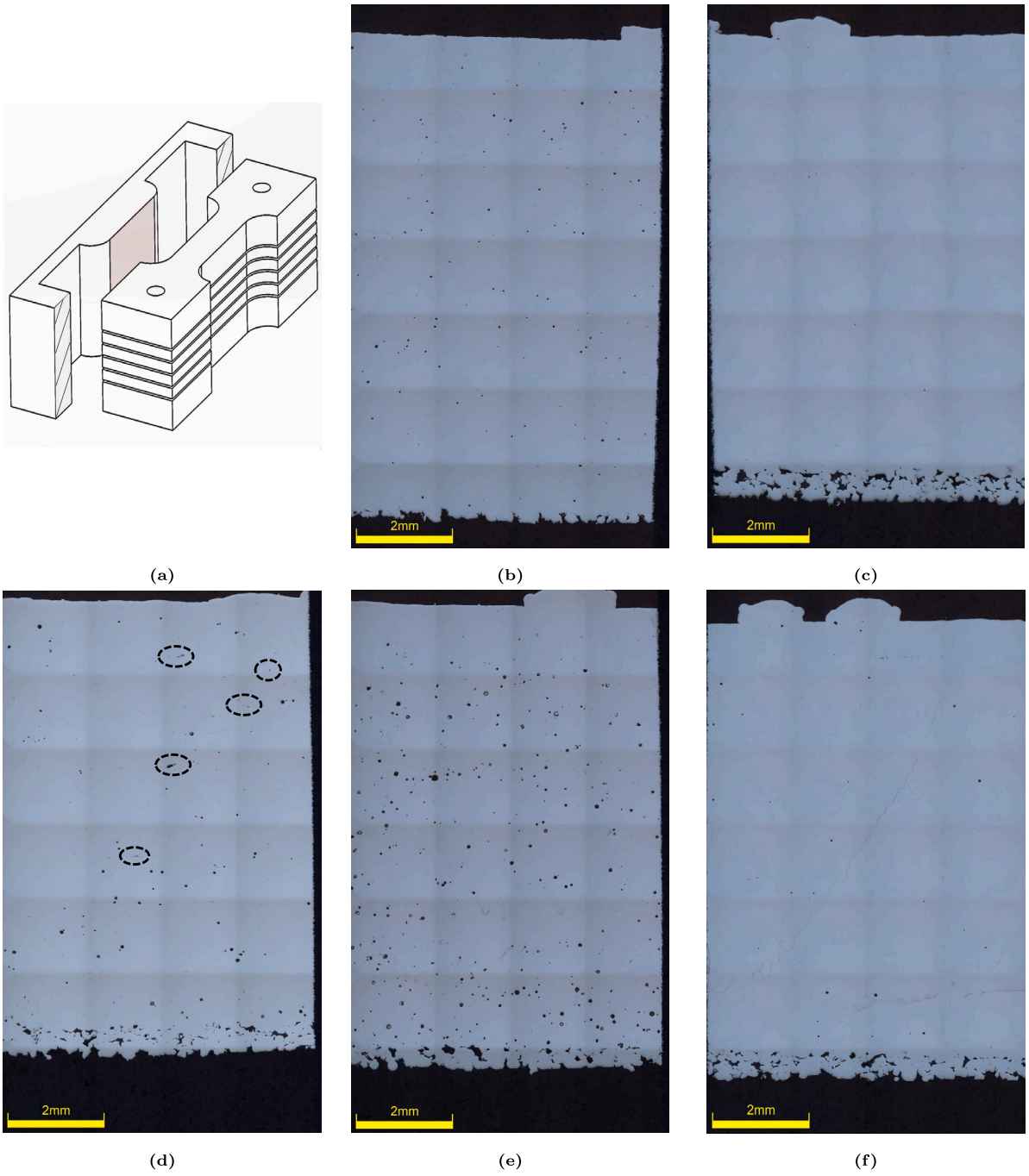


Fig. 14. Overall OM porosity of the selected parts where (a) represents the sub-sized samples cut from the printed block, (b), (c), (d), (e) and (f), are examined using optical microscopy and corrected to view the optical density of the samples calculated by (1- area porosity%) as 99.83%, 99.99%, 99.74%, *99.79%* and 99.94%, respectively. Non-spherical (interpreted as lack of fusion) porosities are highlighted in (d).

from the build plate. To determine if porosity was a function of distance from the build plate, cross-sections in a plane perpendicular to the fracture location and vertically aligned with the build direction were used to conduct optical porosity frequency calculations, the results of which can be observed in Fig. 14.

The cross-sections resulted in two observations; the first, is that the lack of fusion porosity, marked by non-symmetrical geometry only occurs in the 90 μm D2 sample, see encircled porosity in Fig. 14(d). It also shows that the 90 μm D3 sample had large amounts of keyholing porosity, Fig. 14(e), but this did not adversely affect the ultimate tensile strength of the material and only had an ≈ 8% reduction in toughness compared to the average of the other 90 μm (excluding D2) specimens.

3.4. Microstructure evolution

To answer whether nickel alloy 718 is mechanically invariant to keyhole porosity, BSE-SEM images were captured using an FEI Quanta 600 to show the microstructure of the samples. The captured BSE images, see Fig. 15 were relatively consistent with the literature on as-built specimens in L-PBF systems; that is, equiaxed grains orienting themselves towards the heat flux from the subsequent layers and dominated by austenitic (γ) and a eutectic mixture of γ and Nb rich laves regions. At high magnifications, MC-type carbides were observed in high frequency. Strengthening circular γ', disk-like γ'', and needle-like δ phases were either not observed or so infrequent as to be considered

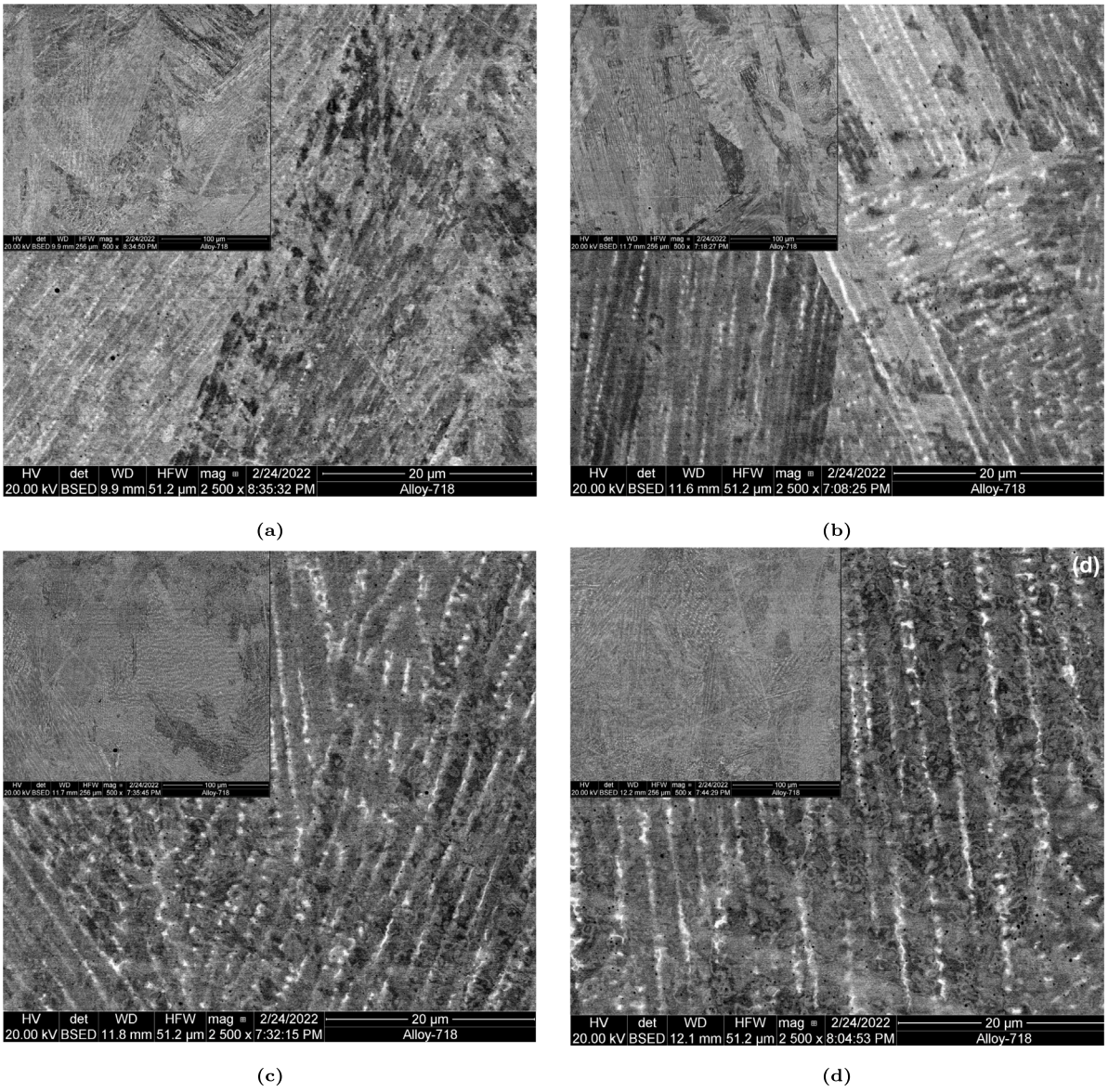


Fig. 15. BSE-SEM images of (a) 60 µm D1, (b) 90 µm D1, (c) 90 µm D2, and (d) 90 µm D3 where the reduced size images are 500× magnification to show larger structure features.

not present. This is consistent with as-built conditions under high cooling rates, which prevent full segregation of Nb, Mo, and C but cannot prevent micro-segregation, which results in the dendritic and cellular structure [26]. Upon heat treatment, the segregated Nb is consumed into the strengthening phases, which gives nickel alloy 718 its unique properties. After this heat treatment, additively manufactured nickel alloy 718 is much more akin to wrought nickel alloy 718, which is rich in γ', γ'', and δ phases [53–55].

4. Conclusion

In this paper, the printability of the nickel alloy 718 is investigated using a systematic framework that provides a thorough search into the parameter space. The framework is applied to two different layer thicknesses to study the effect of the increased layer thickness on the densification and performance of nickel alloy 718 parts.

As the required energy input to provide a large enough melt pool increases significantly for the larger layer thickness, the printability region for the 90 µm layer thickness was reduced significantly as compared to that for the 60 µm layer thickness. The reduced printability region hindered the increase in the scanning speed. Moreover, the larger layer thickness required the use of smaller hatch spacing values. As a result, the maximum achievable VBR for the 90 µm layer thickness is almost equal to that of the 60 µm layer thickness with a slight edge to the smaller layer thickness with values of 7.102 and 6.922 mm^3/s for the 60 and 90 µm layer thicknesses, respectively. It is worth mentioning that the used maximum hatch spacing criterion is a conservative criterion; relaxing it provides higher VBR values. However, for the same VBR, using 90 µm saves on layering time, hence, increasing productivity. Meanwhile, for both layer thicknesses, parts selected from the good printability region were successfully printed with 99.99% OM density.

The printed parts with the highest density were then tested under tension to assess their performance. The samples printed using a 60 µm layer thickness showed, on average, higher ultimate tensile and yield strengths. The average UTS for the 60 and 90 µm layer thicknesses are 1047.3 and 1021.1 MPa, respectively. Nevertheless, the maximum achievable UTS for the 60 µm layer thickness is 1059.3 MPa, while it was 1063.5 MPa for the 90 µm one. Notably, these values are comparable to the values in the literature of 1051 MPa in [27] and 1050 MPa

in [19]. Meanwhile, using an independent-sample t-test showed that the elongation at fracture and the toughness were consistent for both layer thicknesses. Because of the difference in the geometry between the standard test specimens, generally reported in the literature, and the sub-sized samples used in this study, comparing the elongation at fracture results to the literature is not feasible.

The microstructure observed for both layer thicknesses is consistent with the literature for the lower layer thickness samples. The BSE-SEM images show equiaxed grains orienting themselves towards the heat flux. Moreover, they showed the expected γ and the mixture of the γ and Nb-rich laves phases.

CRediT authorship contribution statement

David Shoukr: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peter Morcos:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Tayler Sundermann:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Thomas Dobrowolski:** Writing – review & editing, Resources, Project administration, Conceptualization. **Chad Yates:** Writing – review & editing, Resources, Project administration, Conceptualization. **Jayesh R. Jain:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Raymundo Arróyave:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Ibrahim Karaman:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Alaa Elwany:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge the support of Baker Hughes Co., United States under Contract No. 2000480. Portions of this research were conducted using the advanced computing resources provided by Texas A&M University High-Performance Research Computing. The authors would like to acknowledge the invaluable insights and support from Dr. Nicholas Barta, Dr. Satya Ganti and Dr. Wei Chen at Baker Hughes Co. Finally, the authors would like to acknowledge the support from Dr. Kadri C. Atli during the conduction of the tension tests.

References

[1] W.E. Frazier, Metal additive manufacturing: a review, J. Mater. Eng. Perform. 23 (6) (2014) 1917–1928.
[2] D. Herzog, V. Seyda, E. Wycisk, C. Emmelmann, Additive manufacturing of metals, Acta Mater. 117 (2016) 371–392.
[3] J.P. Oliveira, A. LaLonde, J. Ma, Processing parameters in laser powder bed fusion metal additive manufacturing, Mater. Des. 193 (2020) 108762.
[4] D. Gu, Y. Shen, Effects of processing parameters on consolidation and microstructure of W-Cu components by DMLS, J. Alloys Compd. 473 (1–2) (2009) 107–115.
[5] M. Sadowski, L. Ladani, W. Brindley, J. Romano, Optimizing quality of additively manufactured Inconel 718 using powder bed laser melting process, Addit. Manuf. 11 (2016) 60–70.
[6] Q. Jia, D. Gu, Selective laser melting additive manufacturing of Inconel 718 superalloy parts: Densification, microstructure and properties, J. Alloys Compd. 585 (2014) 713–721.
[7] M. Xia, D. Gu, G. Yu, D. Dai, H. Chen, Q. Shi, Porosity evolution and its thermodynamic mechanism of randomly packed powder-bed during selective laser melting of Inconel 718 alloy, Int. J. Mach. Tools Manuf. 116 (2017) 96–106.
[8] M. Ma, Z. Wang, X. Zeng, Effect of energy input on microstructural evolution of direct laser fabricated IN718 alloy, Mater. Charact. 106 (2015) 420–427.
[9] K. Prashanth, S. Scudino, T. Maity, J. Das, J. Eckert, Is the energy density a reliable parameter for materials synthesis by selective laser melting? Mater. Res. Lett. 5 (6) (2017) 386–390.
[10] R. Seede, D. Shoukr, B. Zhang, A. Whitt, S. Gibbons, P. Flater, A. Elwany, R. Arroyave, I. Karaman, An ultra-high strength martensitic steel fabricated using selective laser melting additive manufacturing: Densification, microstructure, and mechanical properties, Acta Mater. 186 (2020) 199–214.
[11] A. Simchi, H. Pohl, Effects of laser sintering processing parameters on the microstructure and densification of Iron powder, Mater. Sci. Eng. A 359 (1–2) (2003) 119–128.
[12] L. Thijs, F. Verhaeghe, T. Craeghs, J. Van Humbeeck, J.-P. Kruth, A study of the microstructural evolution during selective laser melting of Ti-6Al-4V, Acta Mater. 58 (9) (2010) 3303–3312.
[13] K. Moussaoui, W. Rubio, M. Mousseigne, T. Sultan, F. Rezai, Effects of selective laser melting additive manufacturing parameters of Inconel 718 on porosity, microstructure and mechanical properties, Mater. Sci. Eng. A 735 (2018) 182–190.
[14] V. Popovich, E. Borisov, A. Popovich, V.S. Sufiiarov, D. Masaylo, L. Alzina, Impact of heat treatment on mechanical behaviour of Inconel 718 processed with tailored microstructure by selective laser melting, Mater. Des. 131 (2017) 12–22.
[15] V. Popovich, E. Borisov, A. Popovich, V.S. Sufiiarov, D. Masaylo, L. Alzina, Functionally graded Inconel 718 processed by additive manufacturing: Crystallographic texture, anisotropy of microstructure and mechanical properties, Mater. Des. 114 (2017) 441–449.
[16] API Standard 6ACRA, Age-hardened Nickel-based alloys for oil and gas drilling and production equipment, American Petroleum Institute, Washington, DC, 2015.
[17] B. Zhang, R. Seede, L. Xue, K.C. Atli, C. Zhang, A. Whitt, I. Karaman, R. Arroyave, A. Elwany, An efficient framework for printability assessment in laser powder bed fusion metal additive manufacturing, Addit. Manuf. 46 (2021) 102018.
[18] S. Nayak, S. Mishra, C. Paul, A. Jinoop, K. Bindra, Effect of energy density on laser powder bed fusion built single tracks and thin wall structures with 100 μm preplaced powder layer thickness, Opt. Laser Technol. 125 (2020) 106016.
[19] Q. Nguyen, D. Luu, S. Nai, Z. Zhu, Z. Chen, J. Wei, The role of powder layer thickness on the quality of SLM printed parts, Arch. Civil Mech. Eng. 18 (3) (2018) 948–955.
[20] S. Nayak, S. Mishra, A. Jinoop, C. Paul, K. Bindra, Experimental studies on laser additive manufacturing of inconel-625 structures using powder bed fusion at 100 μm layer thickness, J. Mater. Eng. Perform. 29 (11) (2020) 7636–7647.
[21] A.F. de Souza, K.S. Al-Rubaie, S. Marques, B. Zluhan, E.C. Santos, Effect of laser speed, layer thickness, and part position on the mechanical properties of maraging 300 parts manufactured by selective laser melting, Mater. Sci. Eng. A 767 (2019) 138425.
[22] S. Wang, Y. Liu, W. Shi, B. Qi, J. Yang, F. Zhang, D. Han, Y. Ma, Research on high layer thickness fabricated of 316L by selective laser melting, Materials 10 (9) (2017) 1055.
[23] S. Dadbakhsh, L. Hao, Effect of layer thickness in selective laser melting on microstructure of Al/5 wt.%Fe_2O_3 powder consolidated parts, Sci. World J. 2014 (2014) 1–10.
[24] S.D. Nath, G. Gupta, M. Kearns, O. Gulsoy, S.V. Atre, Effects of layer thickness in laser-powder bed fusion of 420 stainless steel, Rapid Prototy. J. 26 (2020) 1197–1208.
[25] X. Shi, S. Ma, C. Liu, C. Chen, Q. Wu, X. Chen, J. Lu, Performance of high layer thickness in selective laser melting of Ti6Al4V, Materials 9 (12) (2016) 975.
[26] S. Sanchez, P. Smith, Z. Xu, G. Gaspard, C.J. Hyde, W.W. Wits, I.A. Ashcroft, H. Chen, A.T. Clare, Powder bed fusion of Nickel-based superalloys: A review, Int. J. Mach. Tools Manuf. 165 (2021) 103729.
[27] V.S. Sufiiarov, A. Popovich, E. Borisov, I. Polozov, D. Masaylo, A. Orlov, The effect of layer thickness at selective laser melting, Procedia Eng. 174 (2017) 126–134.
[28] P. Paradise, D. Patil, N. Van Handel, S. Temes, A. Saxena, D. Bruce, A. Suder, S. Clonts, M. Shinde, C. Noe, et al., Improving productivity in the laser powder bed fusion of Inconel 718 by increasing layer thickness: Effects on mechanical behavior, J. Mater. Eng. Perform. (2022) 1–16.
[29] V.S. Sufiiarov, E.V. Borisov, I.A. Polozov, Selective laser melting of the Inconel 718 Nickel superalloy, in: Applied Mechanics and Materials, 698, Trans Tech Publ, 2015, pp. 333–338.
[30] B. Sutton, E. Herderick, R. Thodla, M. Ahlfors, A. Ramirez, Heat treatment of alloy 718 made by additive manufacturing for oil and gas applications, JOM 71 (3) (2019) 1134–1143.
[31] J.J. Debarbadillo, S.K. Mannan, Alloy 718 for oilfield applications, JOM 64 (2) (2012) 265–270.

[32] Inconel alloy 718 datasheet, Special Metals Corporation (2007).
[33] E2594-20: Standard Test Method for Analysis of Nickel Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance-Based)., ASTM International West Conshohocken, PA, 2020.
[34] E2823-17: Standard Test Method for Analysis of Nickel Alloys by Inductively Coupled Plasma Mass Spectrometry (Performance-Based)., ASTM International West Conshohocken, PA, 2017.
[35] Metal Powder Industries Federation (MPIF) Standards, Standard Test Method 67, 2022.
[36] G.L. Donati, R.S. Amais, C.B. Williams, Recent advances in inductively coupled plasma optical emission spectrometry, J. Anal. At. Spectrom. 32 (7) (2017) 1283–1296.
[37] D. Beauchemin, Inductively coupled plasma mass spectrometry, Anal. Chem. 80 (12) (2008) 4455–4486.
[38] ASTM International, ASTM B213-20, Standard Test Methods for Flow Rate of Metal Powders Using the Hall Flowmeter Funnel, ASTM International West Conshohocken, PA, USA, 2020.
[39] ASTM B212, Standard Test Method for Apparent Density of Free-Flowing Metal Powders Using the Hall Flowmeter Funnel, American Society for Testing and Materials West Conshohocken, 1999.
[40] T. Eagar, N. Tsai, Temperature fields produced by traveling distributed heat sources, Welding Journal 62 (12) (1983) 346–355.
[41] T. DebRoy, H. Wei, J. Zuback, T. Mukherjee, J. Elmer, J. Milewski, A.M. Beese, A.d. Wilson-Heid, A. De, W. Zhang, Additive manufacturing of metallic components–process, structure and properties, Prog. Mater. Sci. 92 (2018) 112–224.
[42] M. Mahmoudi, G. Tapia, K. Karayagiz, B. Franco, J. Ma, R. Arroyave, I. Karaman, A. Elwany, Multivariate calibration and experimental validation of a 3D finite element thermal model for laser powder bed fusion metal additive manufacturing, Integrat. Mater. Manuf. Innov. 7 (3) (2018) 116–135.
[43] G. Tapia, W. King, L. Johnson, R. Arroyave, I. Karaman, A. Elwany, Uncertainty propagation analysis of computational models in laser powder bed fusion additive manufacturing using polynomial chaos expansions, J. Manuf. Sci. Eng. 140 (12) (2018) 121006.
[44] H. Hosaeus, A. Seifter, E. Kaschnitz, G. Pottlacher, Thermophysical properties of solid and liquid Inconel 718 alloy, High Temp. High Pressures(UK) 33 (4) (2001) 405–410.
[45] K. Shah, I.U. Haq, S.A. Shah, F.U. Khan, M.T. Khan, S. Khan, Experimental study of direct laser deposition of Ti-6Al-4V and Inconel 718 by using pulsed parameters, Sci. World J. 2014 (2014) 1–6.
[46] M.C. Kennedy, A. O'Hagan, Bayesian calibration of computer models, J. R. Stat. Soc. Ser. B Stat. Methodol. 63 (3) (2001) 425–464.
[47] ASTM International, B962-15: Standard test methods for density of compacted or sintered powder metallurgy (PM) products using Archimedes' principle., ASTM International West Conshohocken, PA, 2015.
[48] W.S. Rasband, Imagej, us national institutes of health, bethesda, maryland, usa, 2011, http://imagej.nih.gov/ij/.
[49] A. Kardak, L. Bilich, G. Sinclair, Stress concentration factors for ASTM E8/E8M-15a plate-type specimens for tension testing, J. Test. Eval. 45 (6) (2017) 2294–2298.
[50] I. Koutiri, E. Pessard, P. Peyre, O. Amlou, T. De Terris, Influence of SLM process parameters on the surface finish, porosity rate and fatigue behavior of as-built Inconel 625 parts, J. Mater Process. Technol. 255 (2018) 536–546.
[51] L.S. Moura, G.D. Vittoria, A.H. Gabriel, E.B. Fonseca, L.P. Gabriel, T.J. Webster, É.S. Lopes, A highly accurate methodology for the prediction and correlation of mechanical properties based on the slimness ratio of additively manufactured tensile test specimens, J. Mater. Sci. 55 (22) (2020) 9578–9596.
[52] Z. Wang, K. Guan, M. Gao, X. Li, X. Chen, X. Zeng, The microstructure and mechanical properties of deposited-IN718 by selective laser melting, J. Alloys Compd. 513 (2012) 518–523, http://dx.doi.org/10.1016/j.jallcom.2011.10.107.
[53] K. Amato, S. Gaytan, L. Murr, E. Martinez, P. Shindo, J. Hernandez, S. Collins, F. Medina, Microstructures and mechanical behavior of Inconel 718 fabricated by selective laser melting, Acta Mater. 60 (5) (2012) 2229–2239, http://dx.doi.org/10.1016/j.actamat.2011.12.032.
[54] E. Chlebus, K. Gruber, B. Kuźnicka, J. Kurzac, T. Kurzynowski, Effect of heat treatment on the microstructure and mechanical properties of Inconel 718 processed by selective laser melting, Mater. Sci. Eng. A 639 (2015) 647–655.
[55] X. Li, J. Shi, C. Wang, G. Cao, A. Russell, Z. Zhou, C. Li, G. Chen, Effect of heat treatment on microstructure evolution of Inconel 718 alloy fabricated by selective laser melting, J. Alloys Compd. 764 (2018) 639–649, http://dx.doi.org/10.1016/j.jallcom.2018.06.112.